

stack & queue

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data structure e.g. array, linked list, (binary) tree, stack, queue

stack and queue let you insert and remove items **at the ends only**

stack

As an ADT(**abstract data type**), insertions and removals of items on a stack are based on the last-in **first-out** (LIFO) policy.

It supports:

- **Data**: an **ordered** list of data/items.
- **Operations** (major ones):
 - top** : **get** the value of the **top** item
 - push** : **add** a new item to the **top**
 - pop** : **remove** an item from the **top**

queue

As an ADT, insertions and removals of items on a queue are based on a **first-in first-out** (FIFO) policy

- **Data**: an **ordered** list of data/items.
 - **Operations** (major ones):
 - front** : **get** the value of the **front** item
- enqueue** : **add** a new item to the **back**
dequeue : **remove** an item from the **front**